Chapter 2
Business Strategy and Frameworks

Quote

"*Strategy without tactics is the slowest route to victory. Tactics without strategy is the noise before defeat."*
Sun Tzu (544 BC - 496 BC), Chinese Military Commander and Author

In this chapter we help the data scientist develop the language business uses. This will allow you, the data scientist, as called out in "Chapter 1", bridge the gap between you and senior executives.

Understanding the Business - Strategy

Mission, Vision, Strategy and Tactics are simple and straightforward concepts, don't overthink them!

- Our MISSION explains why we exist
- Our VISION is a goal – It is NOT the same as our STRATEGY; our VISION is the targeted end-result
- Our STRATEGY is a high level plan; our STRATEGY tells us how we are going to achieve (or sustain) our VISION
- TACTICS are the specific timed steps we'll take to achieve our STRATEGY
- Our CORE VALUES, or guiding principles, are the fundamental beliefs that form our moral compass and drives our corporate culture; they dictate how we interact with customers, suppliers, colleagues and other stakeholders

Everyone in the organization should know why they exist, what they hope to achieve and how they're going to get there. Corporations don't always provide a consistent view of their mission, vision, strategy and core values within their larger competitive and economic environment. Overtime organizations change, and sometime must change, their strategy based on market conditions. When an organizations strategy is out of sync with their environment, corporations often falter and need to evaluate their strategy. Organizations often seek the help of management consulting firms (e.g. McKinsey, Boston Consulting Group) to clarify their strategy (and sometimes their mission, vision and core values). Many startups, and at times larger organizations, "pivot" to more fully leverage their existing capabilities within their ecosystem. A pivot is nothing more than a redefinition of one or more of the organization's mission, vision, strategy and core values. Although organizations should routinely evaluation their strategy, at times "disruption" forces an organization to quickly respond.

We'll explore the mission, vision, strategy, core values of some well-known corporations to provide perspective on these terms.

Sample #1: Walmart	
Mission	To help people save money so they can live better
Vison	To become the worldwide leader in retailing
Strategy	Price, access, assortment & experience drive a customer's choice of retailer. Historically, Walmart led on price and assortment. Retail environments are more competitive today, especially with e-commerce. To win, Walmart will lead on price, invest to differentiate on access, be competitive on assortment & deliver a great experience.
Core Values	**Service to the Customer** • Customer First– Listen to, anticipate and serve customer wants and needs • Frontline Focused – Support and empower associates to serve customer everyday • Innovative and Agile – Be creative, take smart risks and move with speed **Respect for the Individual** • Listen– Be visible and available; collaborate with others and be open to feedback

	• Lead by Example– Be humble, teach and trust others to do their jobs; give honest and direct feedback • Inclusive– Seek and embrace differences in people, ideas, and experiences **Strive for Excellence** • High performance – Set and achieve aggressive goals • Accountable – Take ownership, celebrate successes and be responsible for results • Strategic – Make clear choices, anticipate changing conditions and plan for the future **Act with Integrity** • Honest – Tell the truth, keep your promises and be trustworthy • Fair – Do right by others; be open and transparent • Courageous – Speak up, ask for help, make tough calls and say no when appropriate
Sample #2: Toyota	
Mission	Toyota seeks to create a more prosperous society through automotive manufacturing
Vison	Toyota aims to achieve long-term, stable growth in harmony with the environment, the global economy, the local communities it serves, and its stakeholders
Strategy	(Business Unit Strategy – US) Focus on designing brands that excite US buyers Be fast and move quickly No more executive rotation "Always make better cars" mentality
Core Values	Guiding Principles at Toyota (http://www.toyota-global.com/company/vision_philosophy/guiding_principles.html) • Honor the language and spirit of the law of every nation and undertake open and fair business activities to be a good corporate citizen of the world. • Respect the culture and customs of every nation and contribute to economic and social development through corporate activities in their respective communities. • Dedicate our business to providing clean and safe products and to enhancing the quality of life everywhere through all of our activities. • Create and develop advanced technologies and provide outstanding products and services that fulfill the needs of customers worldwide. • Foster a corporate culture that enhances both individual creativity and the value of teamwork, while honoring mutual trust and respect between labor and management. • Pursue growth through harmony with the global community via innovative management. • Work with business partners in research and manufacture to achieve stable, long-term growth and mutual benefits, while keeping ourselves open to new partnerships. Established in 1992, revised in 1997. (Translation from original Japanese) Five Main Principles of Toyoda - Sakichi Toyoda, Founder of Toyota • Always be faithful to your duties, thereby contributing to the company and to the overall good. • Always be studious and creative, striving to stay ahead of the times. • Always be practical and avoid frivolousness. • Always strive to build a homelike atmosphere at work that is warm and friendly. • Always have respect for spiritual matters, and remember to be grateful at all times.
Sample #3: HSBC	
Mission	Through an international network linked by advanced technology, including a rapidly growing e-commerce capability, HSBC provides a comprehensive range of financial services: personal financial services; commercial banking; corporate, investment banking and markets; private banking; and other activities
Vison	We aspire to be one of the world's great specialist banking groups, driven by commitment to our core philosophies and values
Strategy	**Our long-term two-part strategy is to:** • Develop our international network. To facilitate international trade and capital flows and serve our clients, with potential to help them grow from small enterprises into large multinationals. • Invest in wealth and retail businesses with local scale. To make the most of global social mobility, wealth creation and long-term demographic changes in our priority markets.
Core Values	**HSBC Core Values (http://www.hsbc.com/our-approach/our-values)** **Dependable** • Standing firm for what is right, delivering on commitments, being resilient and trustworthy • Taking personal accountability, being decisive, using judgment and common sense, empowering others **Open to different ideas and cultures** • Communicating openly, honestly and transparently, welcoming challenge, learning from mistakes • Listening, treating people fairly, being inclusive, valuing different perspectives **Connected to customers, communities, regulators and each other** • Building connections, being aware of external issues, collaborating across boundaries • Caring about individuals and their progress, showing respect, being supportive and responsive

At a high level, mission, vision, strategy and core values should inform the data scientist on the applicability of the analytics that make sense to the organization. For example, in thriving economic times, strategy is often focused on growth. Analytics will likely need to provide growth opportunities by identifying, for example new markets or new products. In tough economic times, strategy is often focused on improving efficiency and likewise analytics will likely follow, by identifying process improvements or cost reductions.

Understanding Business Value – Framing the Problem

Fundamentally, data scientists are NOT solving analytics problems, they must use analytics to solve business problems. Businesses are constantly dealing with constrained optimizations. In order to understand issues or answer business questions, organizations leverage business frameworks. No matter what an organization does (e.g. sell products, deliver services, sell data), they draw upon frameworks to help understand specific situations. Frameworks typically provide one of the following focuses:

- Strategic
- Economic
- Financial

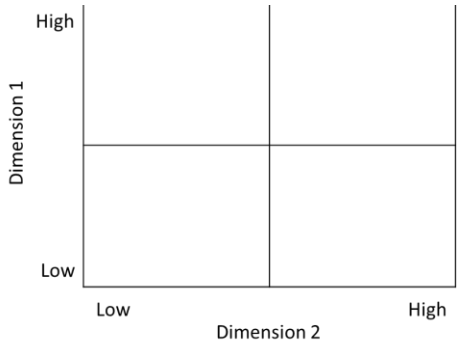
As described in Chapter 1, for the business, and especially senior executives, understanding how analytics solve a business issue must be communicated in the "language" that the business understands. The data scientist can leverage business frameworks to simplify this communication. The choice of the framework is based on the specific business issue or question that needs to be answered. Familiarity and experience with the frameworks are required in order to quickly choose the one, two, or potentially a few frameworks that help in the given situation. There are many resources available to help build your knowledge of business frameworks and help you recognize the relevance of business frameworks (see "Additional Resources" at the end of the chapter for a listing) However, as is often quoted, "practice makes perfect" – the data scientist must use the frameworks in order to develop a full understanding.

Framework Overview

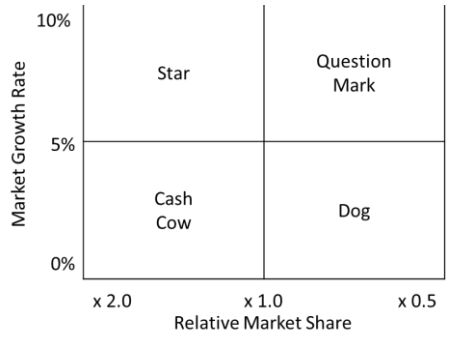
Frameworks help evaluate strategic, economic and financial tradeoffs the business must make. Example frameworks include:

- **Value Chain.** A framework for understanding the flow of goods, services and money in an industry. Components include Inbound Logistics, Production Logistics, Outbound Logistics, Sales and Marketing, Service (see Section below for details).
- **Porter's Five Forces.** Developed by HBS professor Michael Porter, this is a framework to analyze an industry. The five forces are Power of Suppliers, Power of Buyers, Substitutes, Rivalry (among firms), and Barriers to Entry (see Section below for more details).
- **SWOT.** A simple presentation framework that allows you to pinpoint an organizations Strengths, Weaknesses, Opportunities, and Threats (see Section below for details).
- **Business Model Canvas.** A business model describes the rationale of how an organization creates, delivers, and captures value. The business model allows you to easily describe and manipulate the business, challenge assumptions, create new strategic alternatives, and innovate. The business model canvas (www.businessmodelgeneration.com) describes the business model through nine basic building blocks that show the logic of how a company intends to make money. The nine blocks cover the four main areas of a business: customers, offer, infrastructure, and financial viability. The business model is like a blueprint for a strategy to be implemented through organizational structures, processes, and systems.

- **3 C's (Company, Customer, Competition).** A framework for analysis that includes understanding the internal operations and strategy of a company, the customers and demand for the product, and competition in existing and new markets.
- **McKinsey 7-S Model.** Strategy (firm's goals), Structure (organizational form that best suits firm's goals), Staffing (what type of staff does firm require), Style (management style), Skills (what skills do employees need), Systems (what measurement systems are needed), Shared Valued (firms' culture)
- **Profitability.** Profit is defined as Revenues - Costs. To analyze a profitability scenario you can analyze revenues (quantity * price) and costs (fixed vs. variable costs discussed above). To fully understand profitability, study an accounting income statement.
- **Supply vs. Demand.** Economic analysis which aims to forecast the price of a good or service based on the intersection of supply and demand curves. Supply analysis looks at production of goods and services to see how much is available for the market. Demand analysis looks at consumers' desire for a good or service and how much they are willing to pay for it. Process Flow
- **Fixed vs. Variable Costs.** Costs can be split between those that stay constant regardless of sales (fixed costs) and those that vary based on the amount of sales (variable costs). An example of a fixed cost is rent. An example of a variable cost is raw materials.
- **Cost vs. Benefit.** A framework that allows you to divide your analysis between the costs that will be incurred by a specific action as well as the benefits that will result from the action
- **4 P's (Product, Price, Promotion, Place).** A framework used to analyze marketing initiatives and new product launches. Product refers to the characteristics of the item being sold. Price refers to the market price. Promotion refers to the sales strategy including PR and Marketing. Place refers to the sales channels product positioning.
- **STP (Segmenting, Targeting, Positioning).** A framework for finding the correct market for a product or service. Segmenting refers to identifying the distinctions between customers. Targeting refers to identifying the target group. Positioning refers to the strategy of appealing to this target group.
- **2 x 2 Matrix.** The 2x2 matrix is a good framework to use any time you have two factors that, when combined, yield different outcomes.



- **Portfolio Management – BCG Growth Share Matrix.** : A 2 x 2 matrix dimensioned by relative market share on the x-axis and market growth rate on the y-axis.



This simple matrix allowed classification of each business unit into a quadrant. Each quadrant having a recommended strategy.

- Stars. Stars operate in high growth industries and have high market share. They generate and use cash. They should be the business units to invest money because Stars typically become Cash Cows and generate positive cash flows. However, not all Stars become Cash Cows. This is especially true in rapidly changing industries where innovative products or services replace existing products. In this case a star can devolve into a dog.
  Strategy: Vertical integration, horizontal integration, market penetration, market development, product development.
- Questions Marks. Question marks require closer consideration. They hold low market share in fast growing markets. They consume cash and incur losses. Question Marks have the potential to gain market share and become a star. Question Marks do not always succeed, even with investment, they can struggle and become dogs.
  Strategy: Market penetration, market development, product development, divestiture
- Cash Cows. Cash cows are the most profitable and should be "milked" to provide as much cash as possible. The cash gained from "cows" should be invested into stars to support their further growth and carefully invested in Question Marks.
  Strategy: Product development, diversification, divestiture, retrenchment.
- Dogs. Dogs have low market share and operate in a slow growing market. They typically are not worth investing given the low or negative returns. Some dogs may be profitable, they may provide needed synergies, or can counter competitor moves.
  Strategy: Retrenchment, divestiture, liquidation

We will now take a deeper dive into a few key frameworks that are often used in business. In "Chapter 4. Analytics Opportunities" we will examine how these frameworks, and others, can be also be used to help identify analytics opportunities.

Value Chain

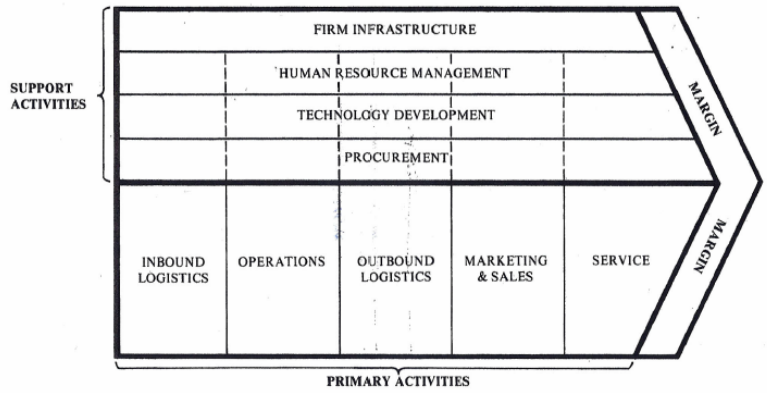
Value Chains take an internal view of the organization and communicates the strategic decisions you have made or could make

View Online Video:
Understanding How Value is Created Within Organizations
https://www.mindtools.com/pages/article/newSTR_66.htm

Michael Porter first discussed Value Chains in his 1985 book, "Competitive Advantage". Understanding how your company creates value, and looking for ways to add value, help in developing a strategy. Porter's Value Chain focuses on systems, and how inputs are changed into the customer consumed outputs. The way in which value chain activities are performed determines costs and affects profits, so this tool can help you understand the sources of value for your organization.

Porter described a chain of common business activities and divided them into primary and support activities.

Porters General Purpose Value Chain



Source: Michael Porter, Competitive Advantage

Your strategy articulates how you will leverage the industry Value Chain and differentiate yourself from competitors as demonstrated by these three competitors:

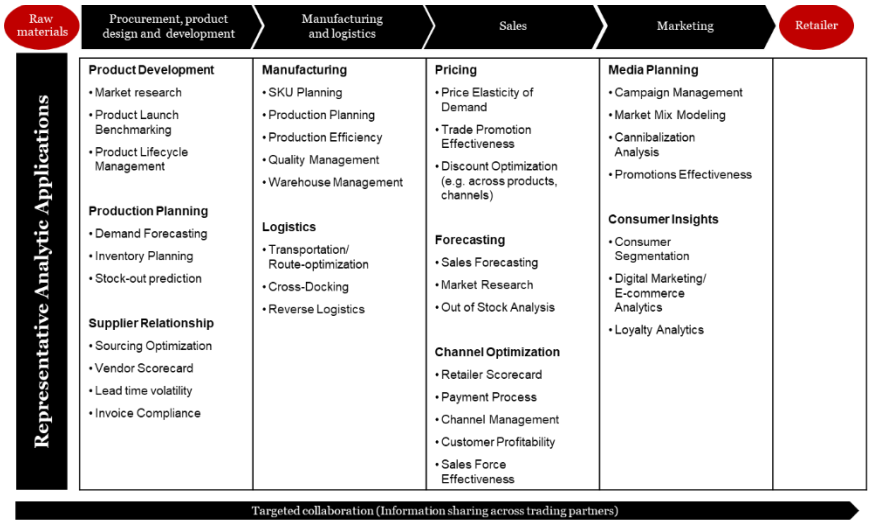
Donated Wheelchairs: Three Competing Value Chains

	Chair design	Operations	Distribution	Provision/ fitting	After-sales repairs
REFURBISHER	NO	Collect & refurbish used chairs	Ship from U.S. to recipients	Send volunteers from U.S.	NO
VOLUME PURCHASER	NO	Outsource production of low-cost chairs	Ship direct from Asian producer to recipients	NO	NO
WHIRLWIND	YES	Partners produce WWI's designs	Regional producers ship to country partners	Local partners do provision & assembly (P&A)	YES P&A centers handle parts & service

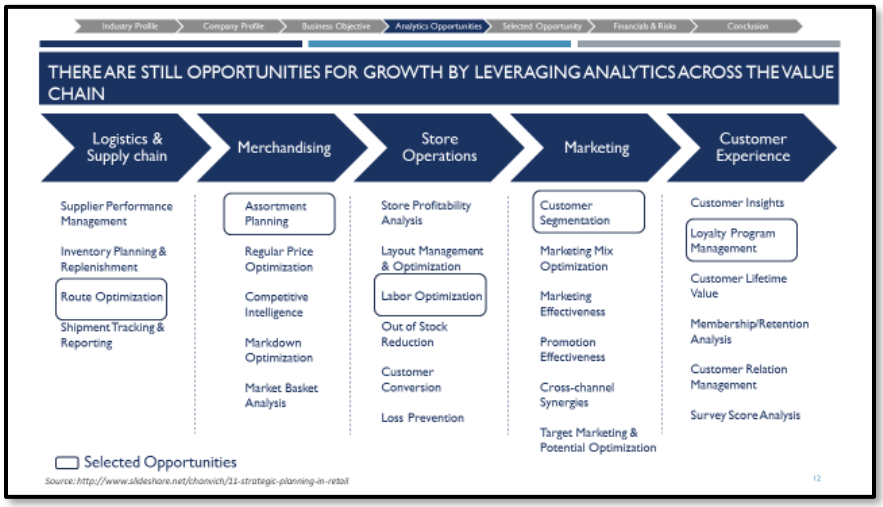
Source: Magretta, The Value Chain and Your P&L.

As a teaser to "Chapter 4: Analytics Opportunities" we share to Value Chains that demonstrate how Analytics can be used to measure key processes in the Value Chain.

Sample #1: Value Chain in the Consumer Packed Goods (CPG) space



Sample #2: Value Chain in a Retail Grocery Store



Porter's Five Forces

The Five Forces framework takes an external view to your strategic positioning

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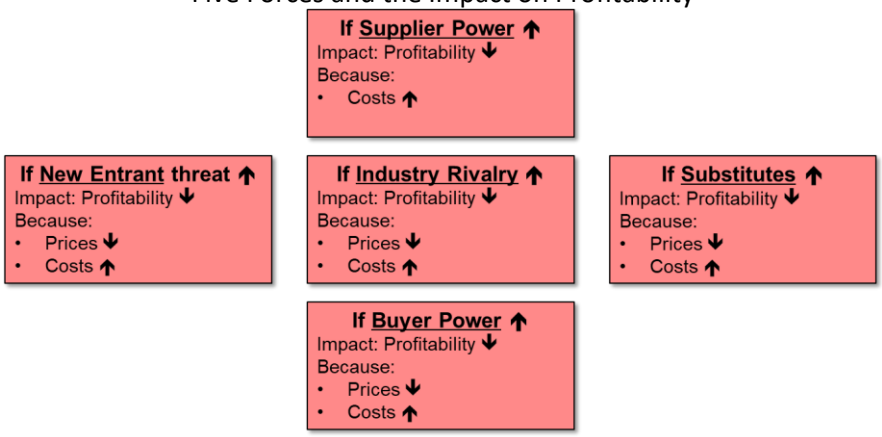
View Online Video
Understanding Competitive Forces to Maximize Profitability
https://www.mindtools.com/pages/article/newTMC_08.htm

Michael Porter created the Five Forces framework to analyze an industry's attractiveness and potential profitability. It is one of the most prevalent business strategy tools. Porter recognized that organizations needed to watch, not only their rivals, but examine the impact of other factors on the business ecosystem. Porter identified five forces that make up the external ecosystem and impact profitability.

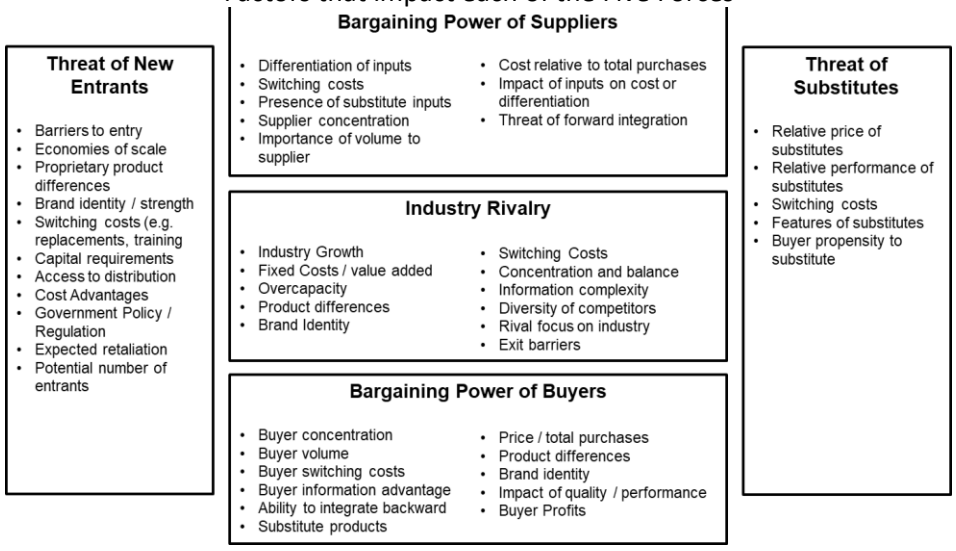
- **Competitive Rivalry.** How many competitors do you have? How strong are they? Who are they? How do their products and services compare with yours? With intense competition, companies can attract customers with aggressive price cuts and impact your profit. In markets with many competitors, suppliers and buyers can go somewhere else to get a better deal. Where competition is low you have strength and higher profits.

- **Supplier Power.** How easy it is for your suppliers to increase their prices? How many potential suppliers do you have? How unique is the product or service they provide? How expensive would switching suppliers be? The more supplier choices, the less power each has and the easier it will be to switch and reduce your costs. With few suppliers, they have stronger position and they can charge more.
- **Buyer Power.** How easy can buyers drive down your prices? How many buyers are there? How large are buyer orders? How much cost would they incur if they switched to a competitor? Can your buyers dictate terms? When you deal with a few customers, they have more power. Your power increases if you have many customers.
- **Threat of Substitution.** Can customers find a different way of achieving what your products or services provide? For example, if you're a restaurant, your customers could choose to eat at home. A substitution that is easier and cheaper can negatively impact your profitability.
- **Threat of New Entry.** Is it easy for someone to enter the market? How much upfront to enter? How much regulation? If it easy to enter the market, you have a weaker position. If there are barriers to entry, then you have a stronger position to maintain profitability.

Five Forces and the Impact on Profitability

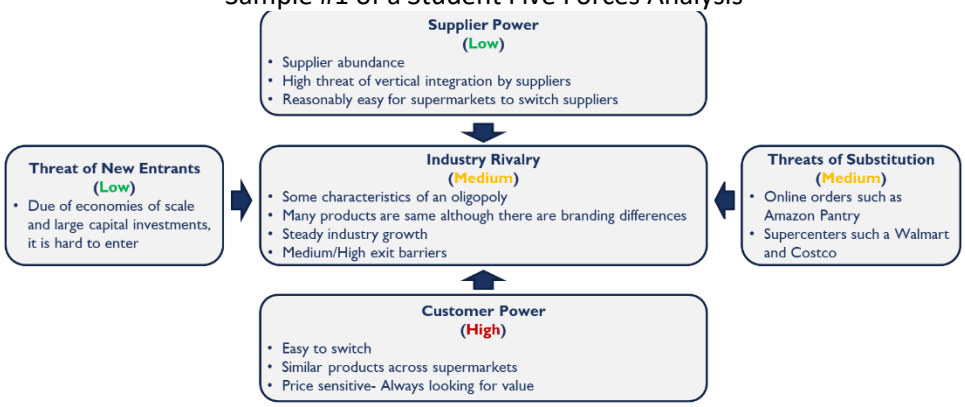


Factors that impact each of the Five Forces



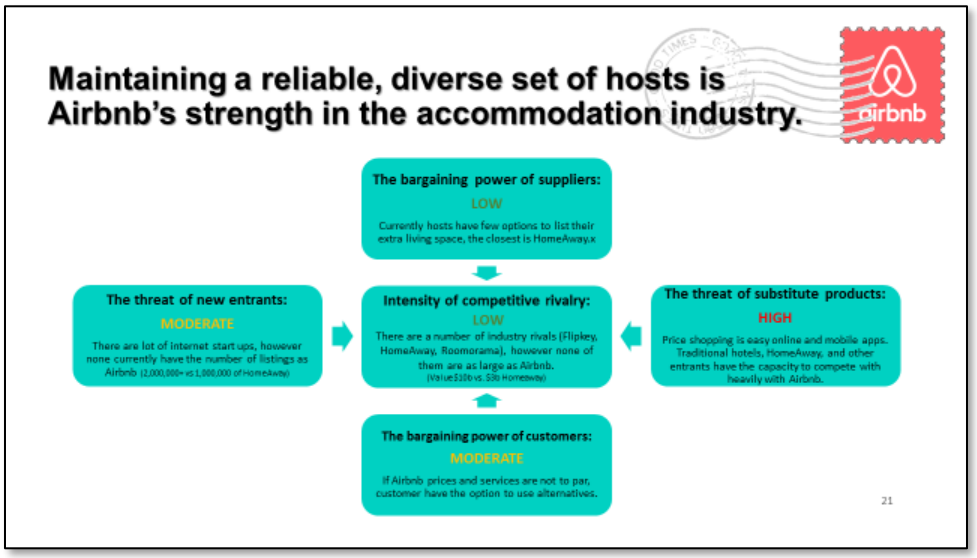
Analytics can be used on attributes of the Five Forces to evaluate areas of differentiation

Sample #1 of a Student Five Forces Analysis



Customer power is high within the supermarket industry

Sample #2 of a Student Five Forces Analysis



Strength Weaknesses Opportunities Threats (SWOT) Analysis

SWOT analysis can be used to understand your strategic position and identify analytics opportunities

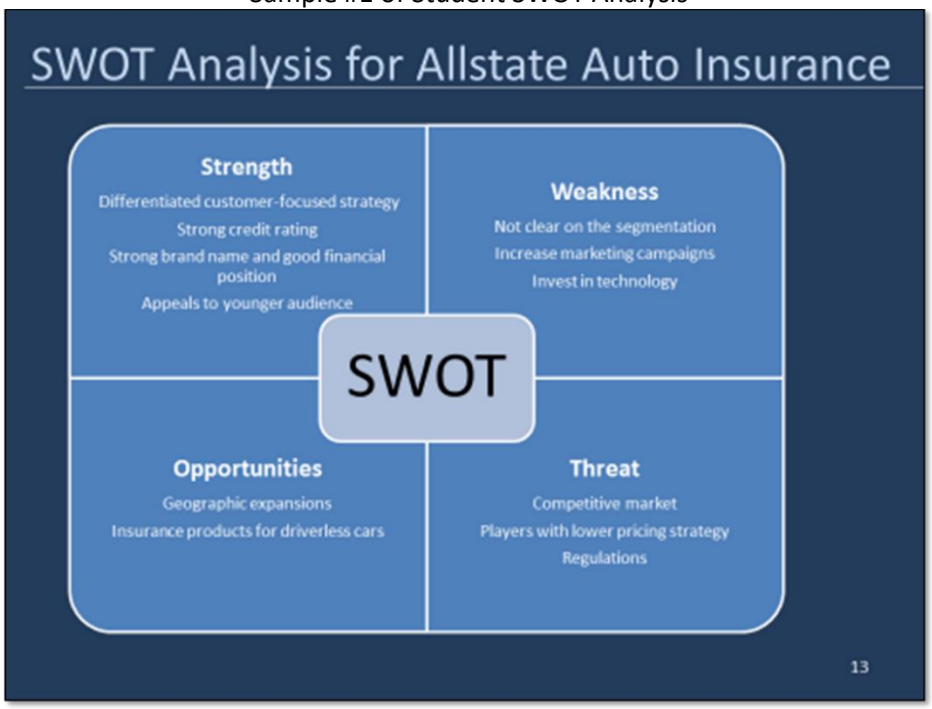
In a SWOT analysis, you consider both inside and outside factors to clarify the world in which your business operates. The SWOT analysis includes your company's internal strengths and weaknesses. These are the under your control. Once you've looked inward to evaluate your strengths and weaknesses, you then scan the outer world for opportunities and threats.

The following provides examples for each of the quadrants of the SWOT analysis.

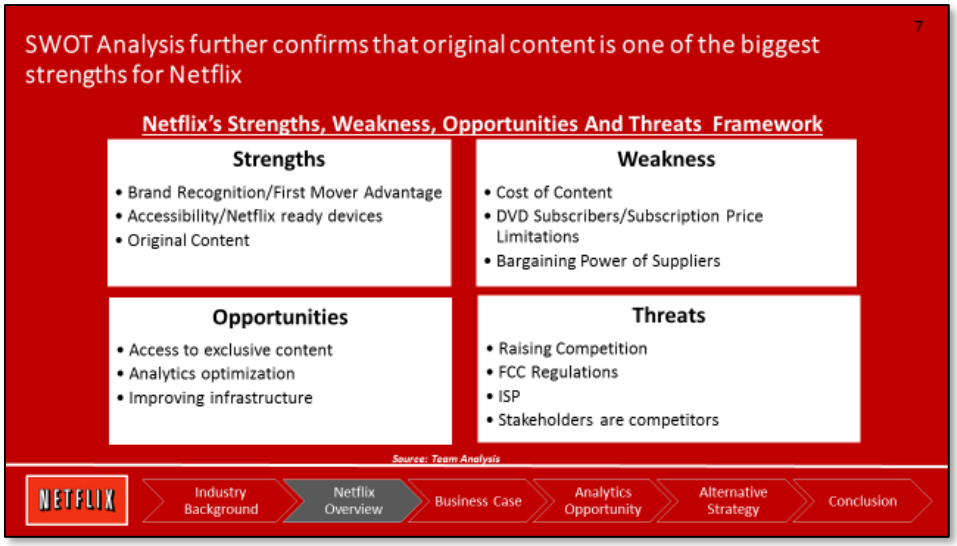
	POSITIVE	NEGATIVE
INTERNAL	**STRENGTHS** • Leadership abilities • Decision-making abilities • Quality • Customer service • Efficiency • Technology • Alliances or partnerships • Intellectual property • Brand • Cost advantages	**WEAKNESSES** • Low quality • Poor customer service • Inefficiency • Lack of technology • No alliances or partnerships • Lack of intellectual property • Weak brand • Poor reputation among customers • High cost structure
EXTERNAL	**OPPORTUNITIES** • Customers • Pricing constraints • Competitor behavior • Changes in technology • Macroeconomic trends • Regulation • Suppliers • Opening of new markets • Related products or services • Potential partners	**THREATS** • A competitor lowering its prices • Shifts in consumer tastes • Increased trade barriers • Legislation or regulations that will increase costs or eliminate a product • A declining product or market • A supplier who is exerting price pressure • A new product or service that could be a substitute for our product or service

Source: HBR, SWOT Analysis Toolkit

Sample #1 of Student SWOT Analysis

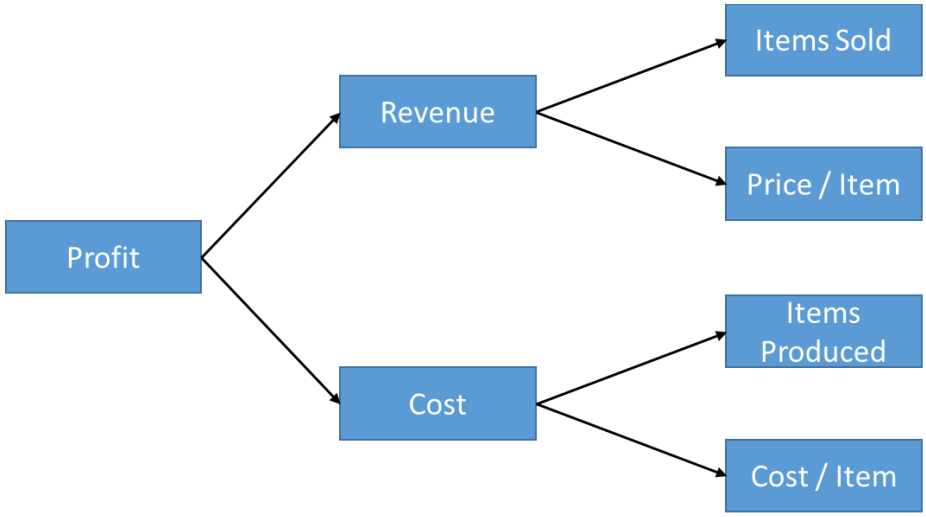


Sample #2 of Student SWOT Analysis



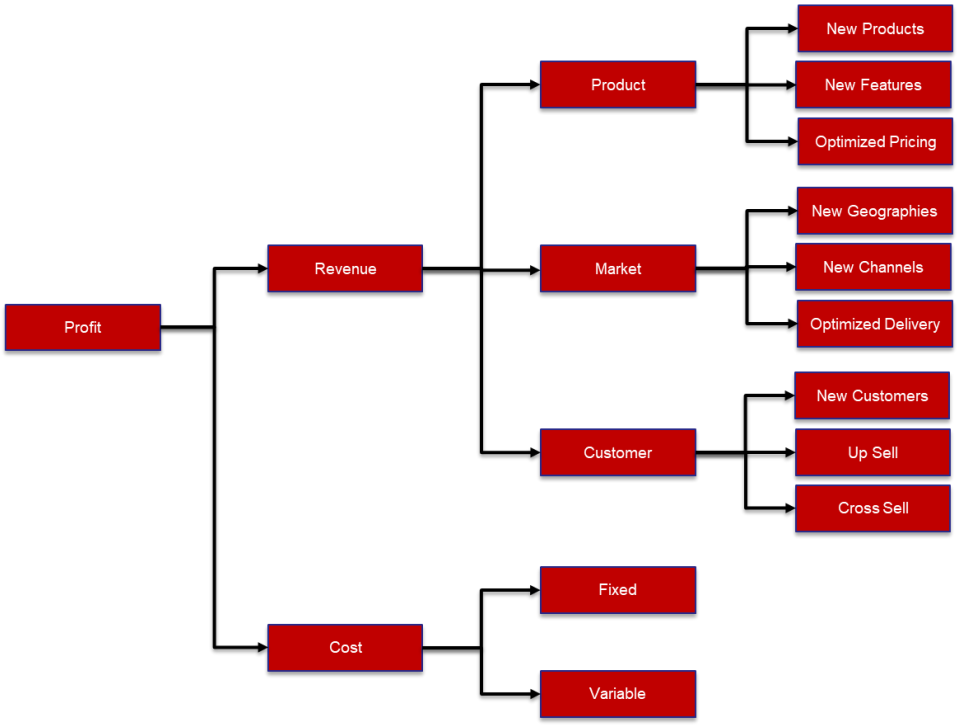
Driver Trees.

A Driver Tree is a representation of a business model that links a business value to a set of drivers. For example, senior executives and stakeholders care about Profit and a first level set of drivers would be Revenue and Cost.



Simple Profit Tree

Driver trees should be detailed to a point that they describe the business value and underlying drivers to a level that is useful for the purpose intended.



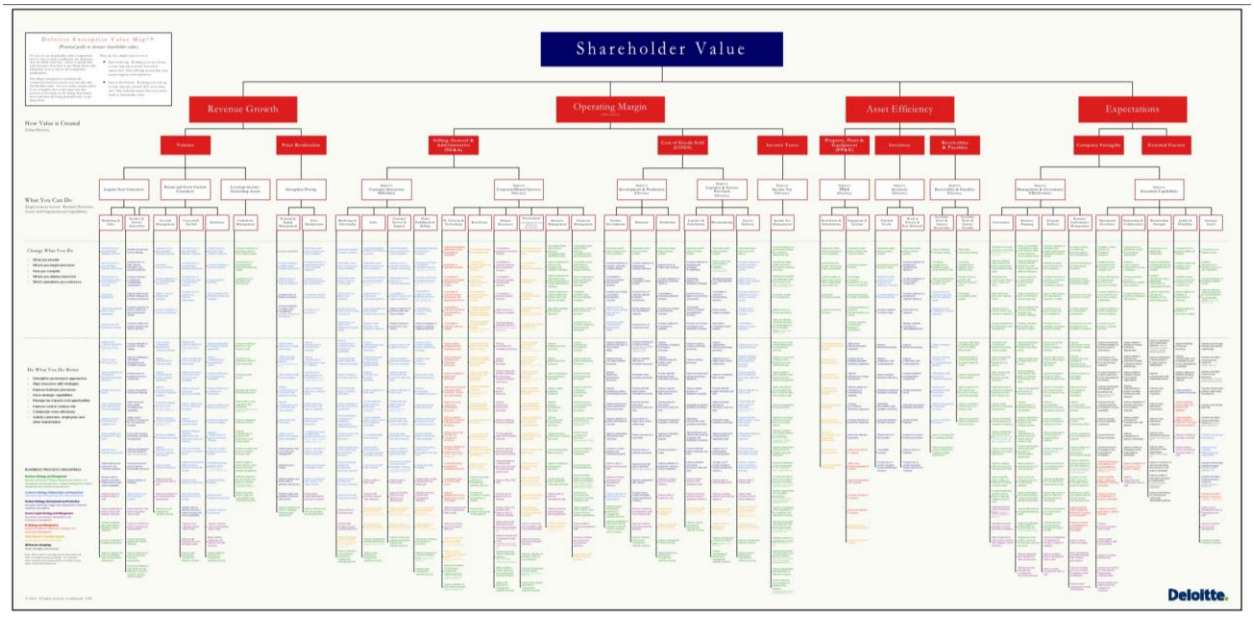
Extended profit-based Driver Tree

Driver Trees were first used by DuPont in the 1920s to describe Return on Investment (ROI). The model included profitability, asset turnover and financial leverage as the initial set of drivers. Each of these were broken down further. The driver tree helped Dupont understand the impact of changes to key business parameters on the business goal of ROI. The Driver Tree was considered revolutionary, not because of the visual representation, but because it allowed business value to be defined in terms of parameters from both the income statement and the balance sheet.

Subsequently, value-based management (VBM) and economic value added (EVA) approaches leveraged driver trees. Because of the effort and enterprise focus of these approaches, the resulting driver trees were used mostly by the C-suite typically with support from management consulting firms. However, today's pervasive need to perform "what-if" scenarios and analyses combined with easily obtained data and readily available analytic tools, driver trees have become a standard tool for business departments and teams.

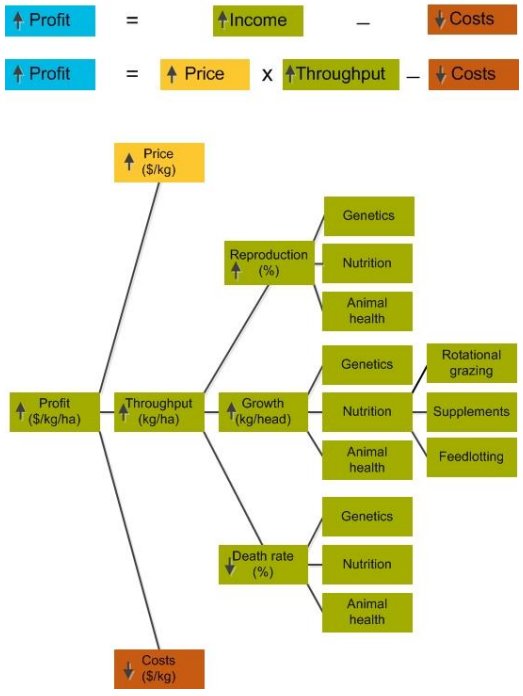
Building a driver tree starts with defining the business goal to be evaluated. At a high-level Return on Investment, Total Shareholder Return, or Profit are typically used. However, driver trees do not have to be financially driven. For example, the business goal could be "Happy Customers." Once the business goal has been defined then the business drivers must be determined. A driver is something the business can influence and has a material impact on the business goal. The driver tree maps how each of these drivers contributes to business goal. The links between the drivers and business goal indicates one thing "leads to" or "affects" the other. Building a driver tree typically requires a whiteboard (physical or electronic) to determine and detail out the linkage of the drivers to the business goal. The final step in developing the tree is to identify which of the drivers are the critical performance drivers – the ones that have the most impact – and focus on them.

The size of a driver tree is determined by the need to understand the drivers of the business goal. It can quite small as long as it provides the insight needed. Although, not typical, a driver tree can be built with EXTREME detail as in this example from Deloitte Consulting.



Driver Trees can be built with ***EXTREME*** detail
Source: Deloitte Consulting

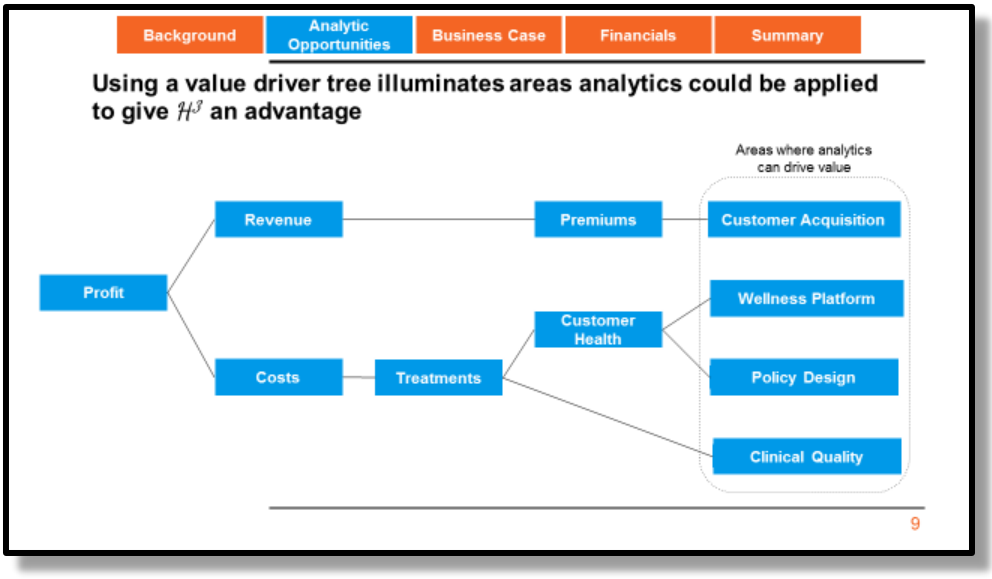
Driver trees can be used across a wide variety of topics and subcomponents created that characterize the intended purpose of the driver tree. In this case to analyze animal production.



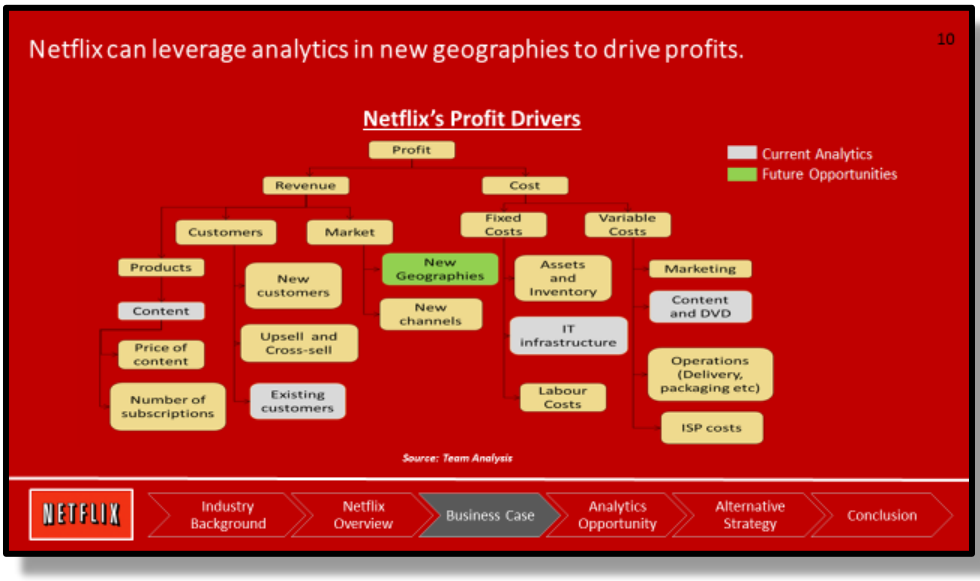
Driver Trees to examine animal production
Source: https://www.agric.wa.gov.au/improvement-tools-driver-trees

Sample Slides

Here are some sample slides from student presentations:



Healthcare Driver Tree



Media Driver Tree

Additional Resources:

Note: The following are paid books and other resources related to business frameworks and how to use them.

Case in Point: Case Interview Preparation, 10 Anniversary Edition, (Marc P. Cosentino, 2009).
http://www.casequestions.com

Crack the Case: How to Conquer Your Case Interviews, (David Ohrvall, 2005).
http://www.mbacase.com

Management Consulting: A Complete Guide to the Industry, (Sugata Biswas and Daryl Twitchell 2002)
http://www.insideconsulting.com/Case%20Interviews.htm

Vault Guide to the Case Interview, (Mark Asher et al., 2007).
http://www.vault.com/wps/portal/usa/store/bookdetail?item_no=848

Ace Your Case II: Mastering the Case Interview by Wetfeet
http://www.cba.pitt.edu/careers/